

Fine-tune the BLIP model for Landmark Recognition Project Report*

*<https://github.com/snuuym/STATS-507/tree/main>

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Abstract—This project explores the fine-tuning of a BLIP (Bootstrapped Language-Image Pretraining) model for recognizing and generating captions for landmark photographs. Leveraging a subset of the Landmark Dataset from Kaggle, the model was trained in a computationally constrained setup. The results demonstrate the model’s capacity for landmark recognition tasks. The link to the code is <https://github.com/snuuym/STATS-507/tree/main>

I. INTRODUCTION

Advancements in image recognition have significantly transformed how we organize and search through visual data. While great strides have been made in recognizing human faces, there remains a noticeable gap in the recognition and categorization of scenic or landmark photographs. This project seeks to address this gap by fine-tuning a pre-trained BLIP model to generate captions for landmark images, thereby enhancing the capability for automated recognition and organization of such visuals.

Image captioning is a prominent area of research, with diverse models and methods evolving to tackle this task. Early approaches relied heavily on static object class libraries, limiting their versatility and adaptability. Subsequent advancements introduced recurrent neural networks (RNNs), which leveraged features from neural networks to better encode sequential dependencies, significantly improving captioning performance (Wang, Zhang & Yu, 2020). [1] This project utilizes the BLIP framework, introduced by Li et al. (2022) [2], which is recognized for its robust generalization capabilities and resilience to noisy pretraining environments (Sun and Min, 2024) [3].

By fine-tuning the BLIP-base model, this work demonstrates how transfer learning can effectively adapt a general-purpose image-captioning framework to a specialized task. This approach not only builds on state-of-the-art methodologies but also underscores the potential of multi-modal models in addressing niche applications.

The primary objectives of this project are:

- 1) To build a model capable of generating accurate captions for images of landmarks.
- 2) To gain hands-on experience in fine-tuning a multi-modal pre-trained model for a specific use case.

II. METHOD

The **methodology** in this project consists of the preparation of data, the selection of a pre-trained model, fine-tuning the

model, and implementing the training and testing process. The goal is to fine-tune a BLIP-base model using a subset of landmark images, enabling it to recognize and generate accurate captions for these landmarks. The methodology is broken down into the following main steps:

A. Dataset Preparation

a) **Dataset Selection:** The Landmark Dataset from Kaggle was used, containing images of 210 different landmarks <https://www.kaggle.com/datasets/andreybeyn/qudata-gemembed-landmarks-210>. Due to computational constraints, the dataset was reduced to 100 landmarks, with 10 images per landmark. The reduced landmark data was chosen according to the alphabetical order. These images were divided into training and testing sets with an 80-20 split. The resulting dataset has 800 training images and 200 testing images.

b) **Dataset Minimization:** The dataset was minimized by selecting the first 100 subfolders (representing 100 landmarks) and limiting the number of images in each subfolder to 10. This was done to balance the dataset’s diversity while adhering to computational constraints. A script was implemented to copy the selected images into a new directory structure for ease of processing.

c) **Train-Test Split:** The data was split into training and testing subsets in an 80:20 ratio. A function was implemented to ensure the training and testing sets were representative of each landmark, preventing potential bias in the learning process. Specifically:

- **Training Set:** 80% of the images used for model fine-tuning.
- **Testing Set:** 20% of the images used for evaluation.

B. Dataset Preprocessing

a) **Image Preprocessing:** Images were preprocessed to standardize inputs and ensure compatibility with the BLIP model’s expectations. The preprocessing steps included:

- Reading the images from the organized directory structure.
- Converting all images into RGB format for uniformity.

b) **Caption Tokenization:** The captions for each image (representing the landmarks’ names) were tokenized using the BLIP model’s processor. Tokenization converts these textual descriptions into numerical sequences that can be used by the model during training and inference.

c) **Dataset Loading:** The processed image paths and their corresponding tokenized labels were loaded into a custom PyTorch Dataset. This enables efficient batch processing during model training.

C. Model Fine-Tuning

The BLIP-base model was chosen for fine-tuning due to its lower computational requirements compared to the original BLIP model, while still maintaining the essential capabilities for image feature extraction and textual tokenization.

a) Training Configuration:

- **Optimizer:** AdamW with a learning rate of $5e-5$.
- **Environment:** The training was conducted in a GPU-enabled Kaggle kernel.
- **Training Duration:** The model was fine-tuned for five epochs.
- **Loss Function:** Loss was computed by comparing the predicted captions with the true labels, guiding parameter updates during backpropagation.

b) **Loss Function:** The model's ability to generate accurate captions was assessed using the **cross-entropy loss**, computed by comparing the predicted sequence of tokens (generated by the model) against the true tokenized captions.

c) **Training Loop:** During each epoch, the following steps were executed in sequence:

- 1) **Forward Propagation:** Input image and tokenized caption data were passed through the model.
- 2) **Loss Computation:** The model's predictions were compared against the true labels using the loss function.
- 3) **Backpropagation:** Gradients were computed to determine how the model's parameters should be updated.
- 4) **Parameter Updates:** Using the computed gradients, weights were updated to minimize the loss.

III. RESULTS

A. Model Setup

The BLIP model and processor were retrieved from Hugging Face and made accessible locally to ensure compatibility with the Kaggle environment. After fine-tuning, the modified model and processor were saved for future use, allowing for reproducibility and further experimentation.

B. Sample Predictions

To evaluate the model's performance, predictions were generated on the test dataset. An example of the results is as follows:

Image: *Blue Mosque in Istanbul Turkey*

Predicted: blue mosque in istabul turkey

True Label: Blue Mosque in Istabul Turkey

Image: *Cloud Gate in Chicago Illinois, USA*

Predicted: la rambla barcelona spain

True Label: "Cloud Gate in Chicago Illinois USA"



Fig. 1. Sample Output.

C. Interpretation of Results

The model successfully demonstrated the ability to recognize and generate captions for many landmarks. However, the accuracy of the generated captions was not entirely satisfactory. Errors were frequently observed, such as the misclassification of Cloud Gate in Chicago as La Rambla in Barcelona. These errors likely stem from two main factors: the limited size of the training dataset and the relatively short training duration of only five epochs, which was implemented to reduce runtime. This indicates opportunities for further improvement, including expanding the training data and optimizing the training loop for better performance.

IV. CONCLUSION

This project fine-tuned a BLIP model for landmark recognition, demonstrating the potential of transfer learning in adapting general-purpose frameworks to address specific tasks. While the results are not entirely satisfactory, they highlight areas for future exploration, such as expanding dataset diversity and refining fine-tuning methodologies to improve generalization. Despite these limitations, the project achieved its subgoal of providing practical experience in fine-tuning models for specific tasks. Completing this project has provided valuable hands-on experience, and therefore, it can be considered a successful completion.

REFERENCES

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